

Recurrence Relations (递推关系)

Rosen 8th ed., §8.1-8.4

8 Advanced Counting Techniques

高级计数技术

- 8.1 Recurrence Relations (递推关系)
- 8.2 Solving Recurrence Relations
 - Linear homogeneous recurrence relations with constant coefficients of degree k (k阶齐次常数线性递推关系)
- 8.3 Divide-and-Conquer Algorithms and Recurrence Relations (分治算法和递推关系)
- 8.4 Generating Functions (生成函数)

§8.1: Recurrence Relations

- A *recurrence relation* (R.R., or just *recurrence*) for a sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more previous elements $a_0, \dots, a_{n-1}$ of the sequence, for all $n \geq n_0$.
 - A recursive definition, without the base cases.
- A particular sequence (described non-recursively) is said to *solve* the given recurrence relation if it is consistent with the definition of the recurrence.
 - A given recurrence relation may have many solutions.

Recurrence Relation Example

- Consider the recurrence relation

$$a_n = 2a_{n-1} - a_{n-2} \quad (n \geq 2).$$

- Which of the following are solutions?

$$a_n = 3n, \quad 2(3(n-1)) - (3(n-2)) = 3n$$

Yes

$$a_n = 2^n, \quad 2 * 2^{n-1} - 2^{n-2}$$

No

$$a_n = 5, \quad 2 * 5 - 5 = 5$$

Yes

Example Applications

- Recurrence relation for growth of a bank account with $P\%$ interest per given period:

$$M_n = M_{n-1} + (P/100)M_{n-1}$$

- $$\begin{aligned} M_n &= M_{n-1} + (P/100)M_{n-1} \\ &= (1 + P/100) M_{n-1} \\ &= r M_{n-1} \quad (\text{let } r = 1 + P/100) \\ &= r (r M_{n-2}) \\ &= r \cdot r \cdot (r M_{n-3}) \quad \dots \text{and so on to } \dots \\ &= r^n M_0 \end{aligned}$$

Example Applications

- Growth of a population in which each organism yields 1 new one every period starting 2 periods after its birth.

$$P_n = P_{n-1} + P_{n-2} \quad (\text{Fibonacci relation})$$

F (n) Rabbits on an island

- $F(0) = 0$
- $F(1) = 1$
- $F(2) = 1$
- $F(3) = 1 + 1 = 2$
- $F(4) = 2 + 1 = 3$
- $F(5) = 3 + 2 = 5 \dots\dots$
- $F(13) = 144 + 89 = 233$

月份	成兔	幼兔	總數
1			1
2			1
3			2
4			3
5			5
6			8
7			13

$$F(0)=0$$

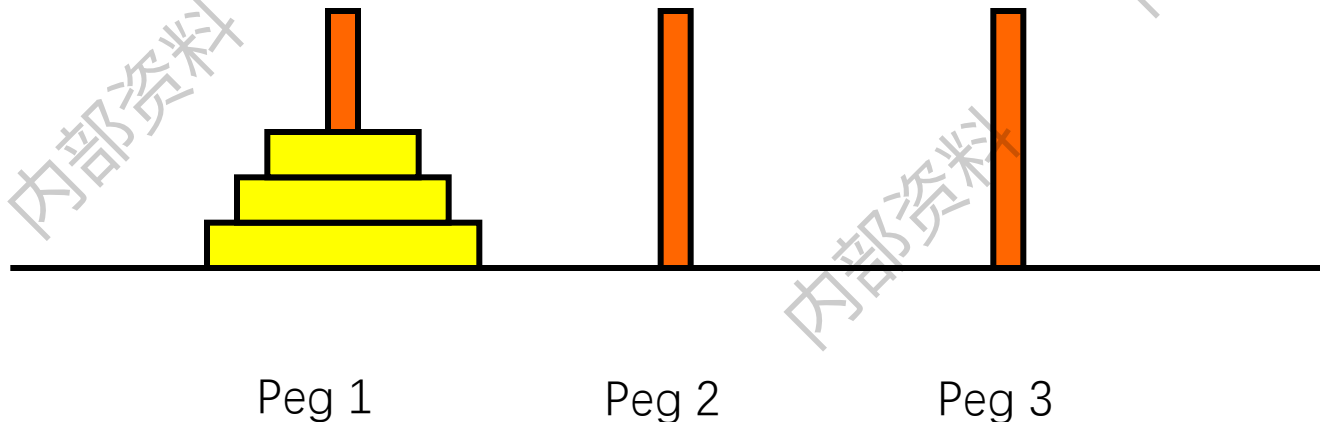
$$F(1)=1$$

$$F(n)=F(n-1)+F(n-2)$$

费波那契递推关系

The Tower of Hanoi Puzzle

- Problem: Get all disks from peg 1 to peg 2.
- Only move 1 disk at a time.
- Never set a larger disk on a smaller one.



Hanoi Recurrence Relation

- Let $H_n = \#$ moves for a stack of n disks.
- Optimal strategy:
 - Move top $n-1$ disks to spare peg. (H_{n-1} moves)
 - Move bottom disk. (1 move)
 - Move top $n-1$ to bottom disk. (H_{n-1} moves)

Note: $H_n = 2H_{n-1} + 1$

Solving Tower of Hanoi RR

$$H_n = 2 H_{n-1} + 1$$

$$= 2 (2 H_{n-2} + 1) + 1 = 2^2 H_{n-2} + 2 + 1$$

$$= 2^2 (2 H_{n-3} + 1) + 2 + 1 = 2^3 H_{n-3} + 2^2 + 2 + 1$$

...

$$= 2^{n-1} H_1 + 2^{n-2} + \dots + 2 + 1$$

$$= 2^{n-1} + 2^{n-2} + \dots + 2 + 1 \quad (\text{since } H_1 = 1)$$

$$= \sum_{i=0}^{n-1} 2^i$$

$$= 2^n - 1 \quad \left(\text{From } \sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r - 1}, \text{ where } a=1, r=2 \right)$$

Backtracking(回溯法)

n=64, 如果每秒移动一次, 需要5849亿年完成

Example 3

- Find a recurrence relation and give initial conditions for the number of bit strings of length N that do not have **two consecutive 0s**. How many such bit strings are there of length **five**?

11**00**10,
01**000**1



11**10**10,
01**010**1



Example 3

Number of bit strings
of length n with no
two consecutive 0s:

End with a 1:

Any bit string of length $n - 1$ with
no two consecutive 0s

1

a_{n-1}

End with a 0:

Any bit string of length $n - 2$ with
no two consecutive 0s

1 0

a_{n-2}

Total:

$$a_n = a_{n-1} + a_{n-2}$$

Bit Strings (*continued*)

$$a_n = a_{n-1} + a_{n-2}$$

The initial conditions are:

- $a_1 = 2$, since both the bit strings 0 and 1 do not have consecutive 0s.
- $a_2 = 3$, since the bit strings 01, 10, and 11 do not have consecutive 0s, while 00 does.

To obtain a_5 , we use the recurrence relation three times to find that:

- $a_3 = a_2 + a_1 = 3 + 2 = 5$
- $a_4 = a_3 + a_2 = 5 + 3 = 8$
- $a_5 = a_4 + a_3 = 8 + 5 = 13$

Note that $\{a_n\}$ satisfies the same recurrence relation as the Fibonacci sequence. Since $a_1 = f_3$ and $a_2 = f_4$, we conclude that $a_n = f_{n+2}$.

Example 4

- A computer system considers a string of decimal digits a valid codeword if it contains an **even number** of 0 digits. For instance, **1230407869** is valid, whereas **120987045608** is **not valid**. Let a_n be the number of valid n -digit codewords. Find a recurrence relation for a_n .

$$a_n = 9a_{n-1} + 10^{n-1} - a_{n-1}$$

Example 4

Case 1 :

The string end with 1~9, the number of valid strings :
 $9 a_{n-1}$

An even number of 0s, a_{n-1}	1~9
---------------------------------	-----

- Case 2:
- The string end with 0, The string of the first $n-1$ digits must be invalid, the number of invalid string is :
 $10^{n-1} - a_{n-1}$,

An odd number of 0s :	0
-----------------------	---

$$10^{n-1} - a_{n-1}$$

Catalan number

Example5

Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n+1$ numbers, $x_0, x_1, x_2, \dots, x_n$, to specify the order of multiplication. For example $C_3=5$ since there are five ways to parenthesize x_0, x_1, x_2, x_3 , to determine the order of multiplication.

$$\begin{aligned} &((x_0 * x_1) * x_2) * x_3 \\ &(x_0 * x_1) * (x_2 * x_3) \end{aligned}$$

$$(x_0 * (x_1 * x_2)) * x_3$$

$$x_0 * ((x_1 * x_2) * x_3)$$

$$x_0 * (x_1 * (x_2 * x_3))$$

Example 5

- 在序列 $x_0 \cdot x_1 \cdot x_2 \cdot \dots x_k \dots x_n$ 中插入有意义的括号，假设有 C_n
- 比如 $((x_0 \cdot x_1) \cdot x_2) \cdot x_3$, $(x_0 \cdot (x_1 \cdot x_2)) \cdot x_3$, 总有一个“ $\cdot$ ”在括号外面。
- 以第 $k+1$ 个“ $\cdot$ ”为分界，

$$\underbrace{x_0 \cdot x_1 \cdot x_2 \cdot \dots x_k}_{C_k} \cdot \underbrace{x_{k+1} \dots \dots x_n}_{C_{n-k-1}}$$

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1}$$

$$C_0 = 1$$

$$C_1 = 1$$

$$C_2 = C_0 C_1 + C_1 C_0 = 2$$

$$C_3 = C_0 C_2 + C_1 C_1 + C_2 C_0 = 5$$

Example 5

- Catalan numbers

The sequence $\{C_n\}$ is the sequence of **Catalan Numbers**. This recurrence relation can be solved using the method of generating functions; see Exercise 41 in Section 8.4.

Catalan 数

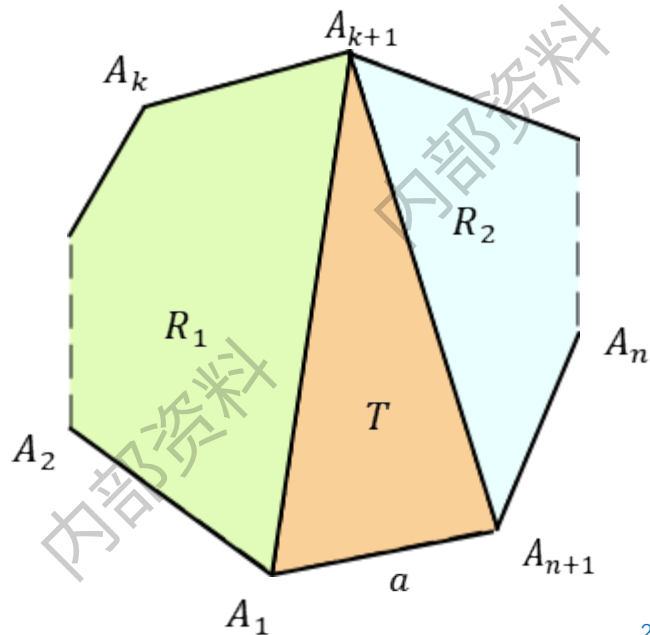
例1：三角划分问题

一个凸 $n+1$ 边形,通过不相交于 $n+1$ 边形的对角线,把 $n+1$ 边形拆分成若干三角形,不同拆分的数目用 h_n 表示.

递推方程

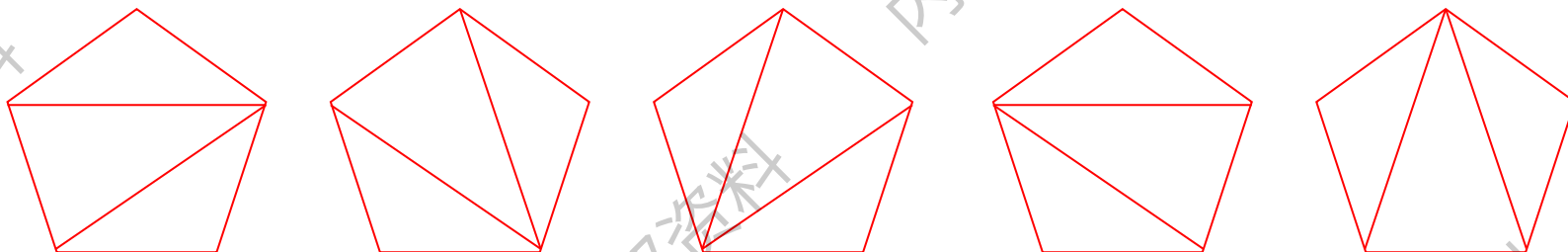
- 考虑 $n+1$ 凸多边形，端点为 A_1, A_{n+1} 的边记为 a ，以 A_{k+1} ($k=1, 2, \dots, n-1$), A_1 和 A_{n+1} 三个点构成三角形 T ，另外 R_1 和 R_2 部分分别构成 $k+1$ 边形和 $n-k+1$ 边形。

$$h_n = \sum_{k=0}^{n-1} h_k h_{n-k} \quad (n \geq 2)$$

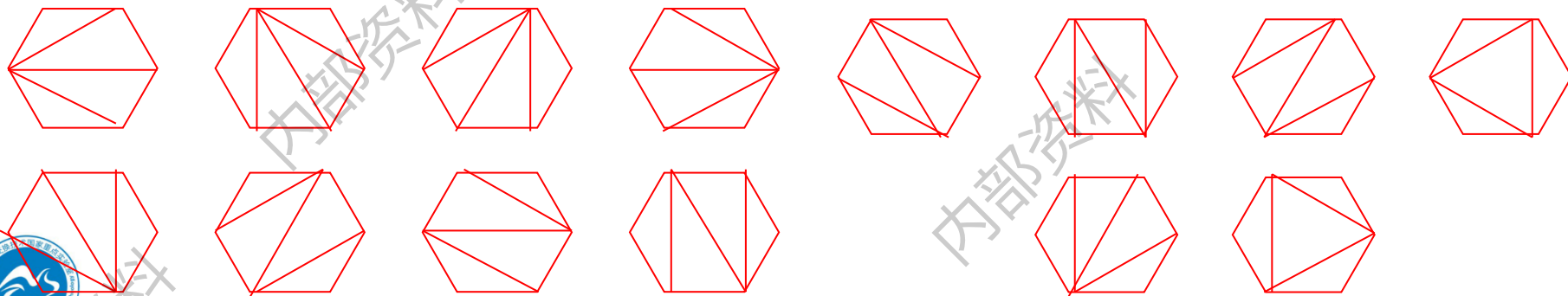


Catalan 数

例如五边形有如下五种拆分方案, $h_4 = 5$



六边形拆分方案: $h_5 = 14$



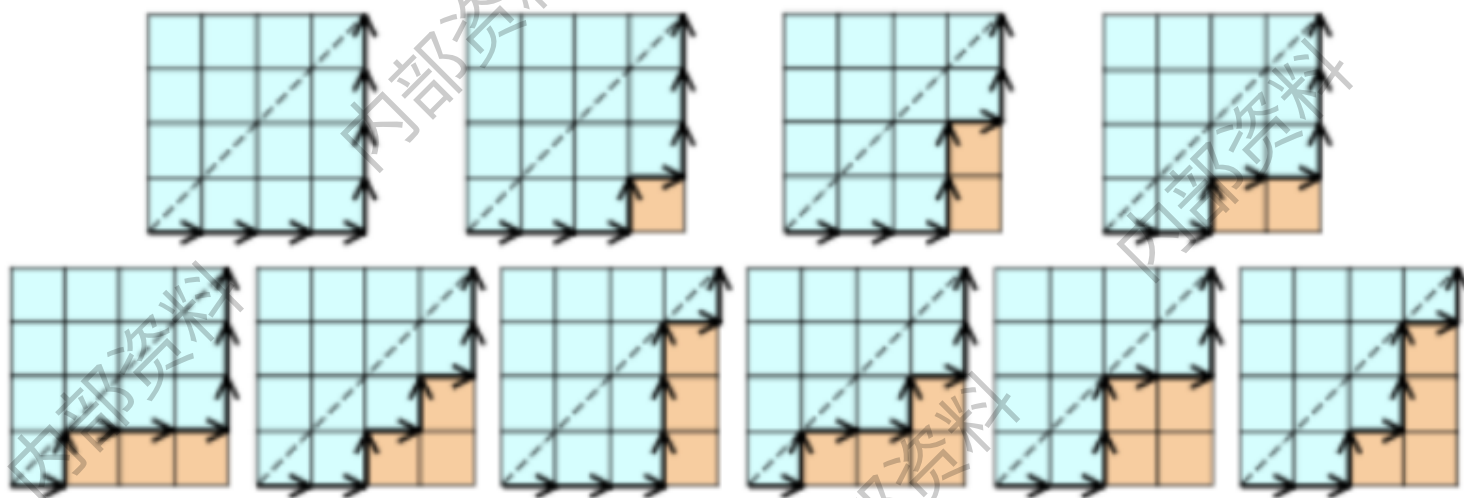
Catalan 数

前16项
Catalan数:

- 1
- 1
- 2
- 5
- 14
- 42
- 132
- 429
- 1430
- 4862
- 16796
- 58786
- 208012
- 742900
- 2774440
- 9694845

Catalan 数

- 一位律师在她住所以北 n 个街区和以东 n 个街区处工作。每天她走 $2n$ 个街区去上班。如果她从不穿越（但可以碰到）从家到办公室的对角线，那么有多少条可能的道路？ H_n



$$(1) H_n = C_{2n}^n - C_{2n}^{n-1}$$

$$(2) H_n = \frac{1}{n+1} C_{2n}^n$$

$$(3) H_n = \frac{4n-2}{n+1} H_{n-1}$$

Catalan 数

- 进栈出栈问题：
饭后，姐姐洗碗，妹妹把姐姐洗过的碗一个
一个放进碗橱摞成一摞(假定碗是有编号的)。
一共有 n 个不同的碗，洗前也是摞成一摞的，
也许因为小妹贪玩而使碗拿进碗橱不及时，
姐姐则把洗过的碗摞在旁边，问：小妹摞起
的碗有多少种可能的方式？ H_n

Catalan 数

• 截止2010年8月

R. Stanley找到了190种不同的问题结构都可以用Catalan数计数。

- Catalan数：以比利时的数学家欧仁·查理·卡特兰（1814 - 1894）命名。
- 清朝数学家明安图（1692年—1763年）在其《割圆密率捷法》中最先发明这种计数方式



Algorithms and Recurrence Relations

算法和递推关系

An algorithm follows the dynamic programming paradigm when it recursively breaks down a problem into simpler overlapping subproblems, and computes the solution using the solutions of the subproblems. Generally, recurrence relations are used to find the overall solution from the solutions of the subproblems. Dynamic programming has been used to solve important problems in such diverse areas as economics, computer vision, speech recognition, artificial intelligence, computer graphics, and bioinformatics.

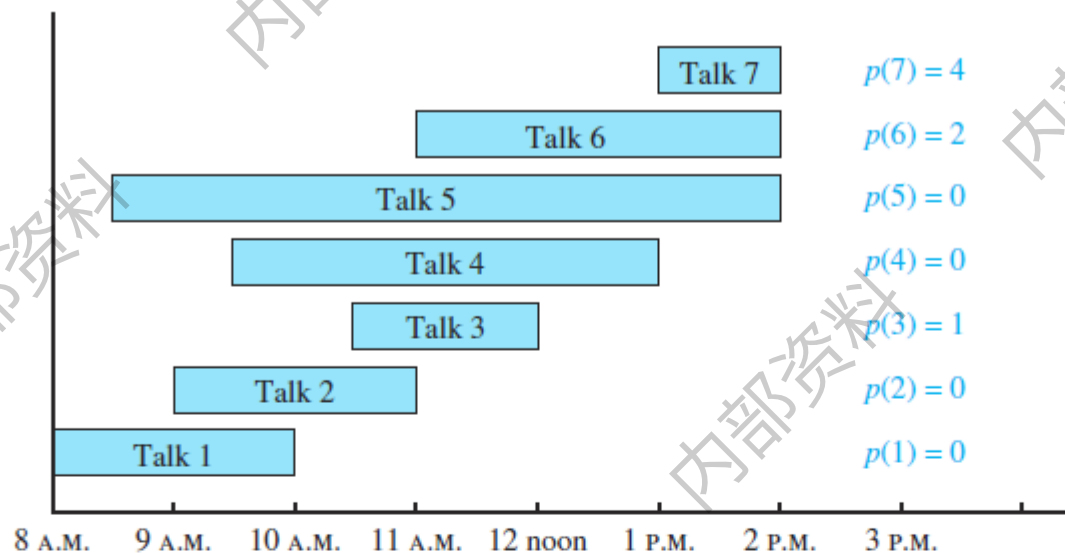


FIGURE 5 A Schedule of Lectures with the Values of $p(n)$ Shown.

EXAMPLE 6 Consider seven talks with these start times and end times, as illustrated in Figure 5.

Talk 1: start 8 A.M., end 10 A.M.

Talk 2: start 9 A.M., end 11 A.M.

Talk 3: start 10:30 A.M., end 12 noon

Talk 4: start 9:30 A.M., end 1 P.M.

Talk 5: start 8:30 A.M., end 2 P.M.

Talk 6: start 11 A.M., end 2 P.M.

Talk 7: start 1 P.M., end 2 P.M.

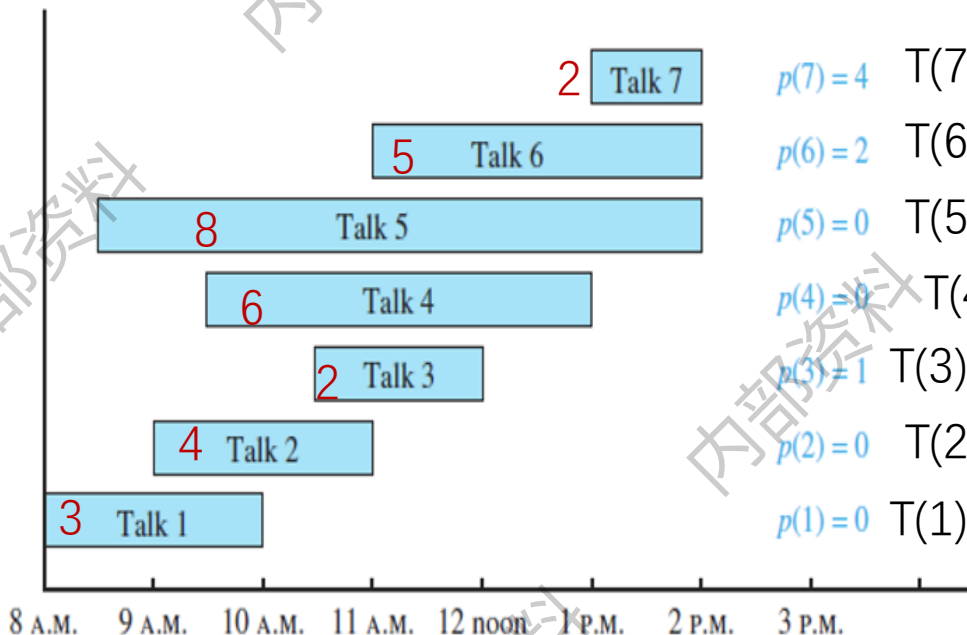
Find $p(j)$ for $j = 1, 2, \dots, 7$.

Solution: We have $p(1) = 0$ and $p(2) = 0$, because no talks end before either of the first two talks begin. We have $p(3) = 1$ because talk 3 and talk 1 are compatible, but talk 3 and talk 2 are not compatible; $p(4) = 0$ because talk 4 is not compatible with any of talks 1, 2, and 3; $p(5) = 0$ because talk 5 is not compatible with any of talks 1, 2, 3, and 4; and $p(6) = 2$ because talk 6 and talk 2 are compatible, but talk 6 is not compatible with any of talks 3, 4, and 5. Finally,

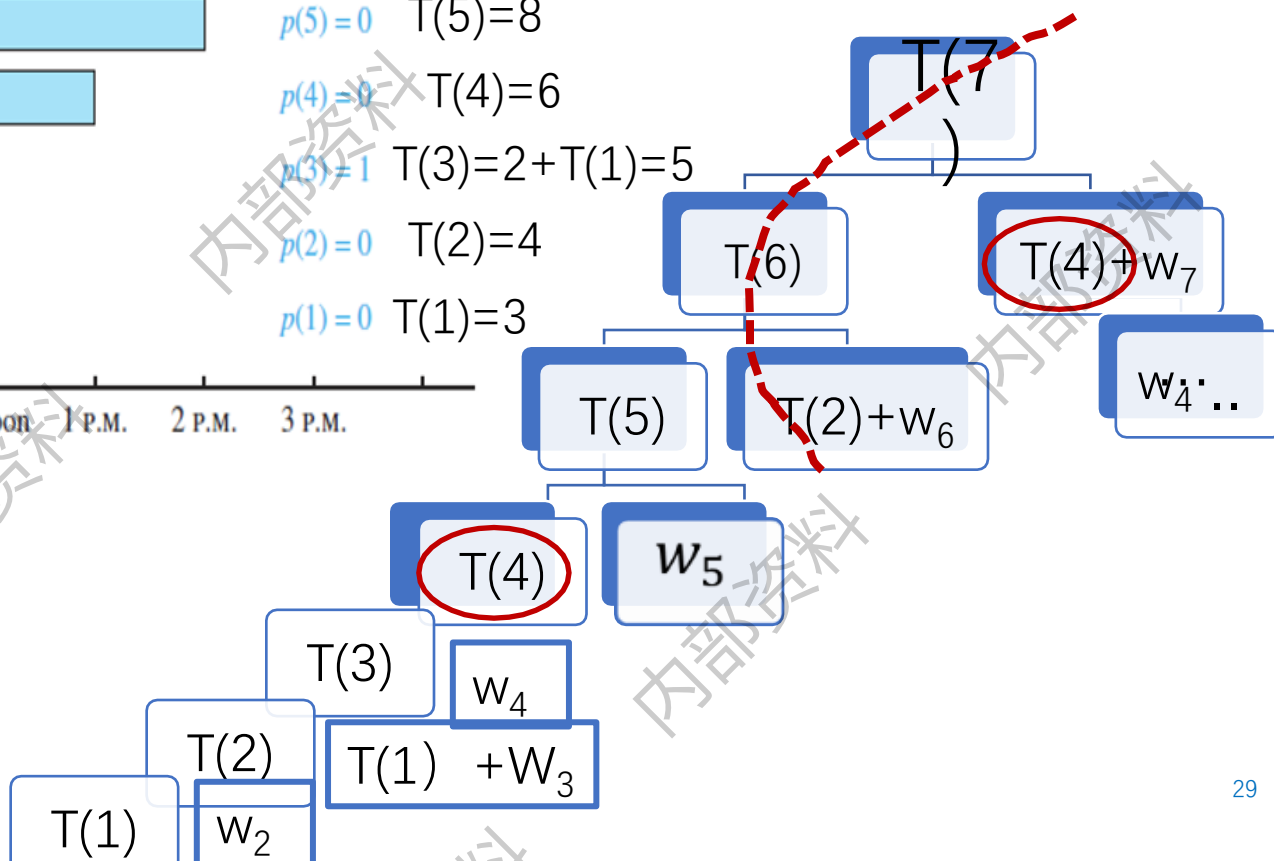
To develop a dynamic programming algorithm for this problem, we first develop a key recurrence relation.

- $T(j) = \max(w_j + T(p(j)), T(j - 1))$.

$$T(i) = \begin{cases} w_i + T(P(i)), & \text{选talk } i \\ T(i-1), & \text{不选talk } i \end{cases}$$



$$\begin{aligned} p(7) &= 4 & T(7) &= T(6) = 9 \\ p(6) &= 2 & T(6) &= 5 + T(2) = 9 \\ p(5) &= 0 & T(5) &= 8 \\ p(4) &= 0 & T(4) &= 6 \\ p(3) &= 1 & T(3) &= 2 + T(1) = 5 \\ p(2) &= 0 & T(2) &= 4 \\ p(1) &= 0 & T(1) &= 3 \end{aligned}$$



§8.2: Solving Linear Recurrences 求解递推关系

- A linear homogeneous recurrence of degree k with constant coefficients (“ k -LiHoReCoCo”)
- k 阶常系数线性齐次递推关系) is a recurrence of the form

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k},$$

where the c_i ($i = 1, \dots, k$) are all real, and $c_k \neq 0$.

- Note
 - On the right-hand side, the summands are each built the same (homogeneous) way as a multiple of one of the k (degree k) previous terms (linear).
 - The solution is uniquely determined if k initial conditions $a_0 \dots a_{k-1}$ are provided.

Examples of Linear Homogeneous Recurrence Relations

- $P_n = (1.21)P_{n-1}$ linear homogeneous recurrence relation of degree one
- $f_n = f_{n-1} + f_{n-2}$ linear homogeneous recurrence relation of degree two
- $a_n = a_{n-1} + a_{n-2}^2$ not linear
- $H_n = 2H_{n-1} + 1$ not homogeneous
- $B_n = nB_{n-1}$ coefficients are not constants

Characteristic equation (特征方程)

- For a linear homogeneous recurrence relation of degree k , $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$
- We call the associated polynomial of degree k , $\chi^k = c_1 \chi^{k-1} + c_2 \chi^{k-2} + \cdots + c_k$, its **characteristic equation (特征方程)**.
- The roots of the characteristic equation play a key role in the explicit formula for the sequence defined by the recurrence relation and the initial conditions.

Solving LiHoReCoCos

- Basic idea: Look for solutions of the form $a_n = r^n$, where r is a constant.
- This requires the *characteristic equation* (特征方程):
$$r^n = c_1 r^{n-1} + \dots + c_k r^{n-k}, \text{ i.e.,}$$
$$r^k - c_1 r^{k-1} - \dots - c_{k-1} r - c_k = 0$$
- The solutions (*characteristic roots*) can yield an explicit formula for the sequence.

特征根与递推方程的关系

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

定理：r 是非零的复数，则 r^n 是递推方程的解
 $\Leftrightarrow r$ 是特征方程特征根

证明： r^n 是递推方程的解

$$\Leftrightarrow r^n - c_1 r^{n-1} - c_2 r^{n-2} - \cdots - c_k r^{n-k} = 0$$

$$\Leftrightarrow r^{n-k} (r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k) = 0$$

$$\Leftrightarrow r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k = 0$$

$$\Leftrightarrow r \text{ 是它的特征根}$$

Solving 2-LiHoReCoCos

- Consider an arbitrary 2-LiHoReCoCo:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

- It has the **characteristic equation** (C.E.):

$$r^2 - c_1 r - c_2 = 0$$

- **Theorem 1:** If this CE has 2 roots $r_1 \neq r_2$, then

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n \text{ for } n \geq 0$$

for some constants α_1, α_2 .

Proof

➤ Suppose that r_1 and r_2 are roots of

- $x^2 - c_1x - c_2 = 0$, so
- $r_1^2 - c_1r_1 - c_2 = 0$,
- $r_2^2 - c_1r_2 - c_2 = 0$ and then
- $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, for $n \geq 1$.

➤ We show that this definition :

- $a_n = c_1 a_{n-1} + c_2 a_{n-2}$
- First we note that α_1 and α_2 are chosen so that $a_1 = \alpha_1 r_1 + \alpha_2 r_2$ and $a_2 = \alpha_1 r_1^2 + \alpha_2 r_2^2$
- and so the initial conditions are satisfied. Then

$$\begin{aligned}r_1^2 - c_1 r_1 - c_2 &= 0, \\r_2^2 - c_1 r_2 - c_2 &= 0\end{aligned}$$

$$\begin{aligned}a_n &= \alpha_1 r_1^n + \alpha_2 r_2^n \\&= \alpha_1 r_1^{n-2} r_1^2 + \alpha_2 r_2^{n-2} r_2^2 \\&= \alpha_1 r_1^{n-2} (c_1 r_1 + c_2) + \alpha_2 r_2^{n-2} (c_1 r_2 + c_2) \\&= c_1 \alpha_1 r_1^{n-1} + c_2 \alpha_1 r_1^{n-2} + c_1 \alpha_2 r_2^{n-1} + c_2 \alpha_2 r_2^{n-2} \\&= c_1 (\alpha_1 r_1^{n-1} + \alpha_2 r_2^{n-1}) + c_2 (\alpha_1 r_1^{n-2} + \alpha_2 r_2^{n-2}) \\&= c_1 a_{n-1} + c_2 a_{n-2}\end{aligned}$$



$$a_0 = C_0 = \alpha_1 + \alpha_2, \quad a_1 = C_1 = \alpha_1 r_1 + \alpha_2 r_2.$$

$$\alpha_2 = C_0 - \alpha_1.$$

$$C_1 = \alpha_1 r_1 + (C_0 - \alpha_1) r_2.$$

$$\text{Hence, } C_1 = \alpha_1 (r_1 - r_2) + C_0 r_2.$$

$$\triangleright \alpha_1 = \frac{C_1 - C_0 r_2}{r_1 - r_2}$$

$$\triangleright \alpha_2 = C_0 - \alpha_1 = C_0 - \frac{C_1 - C_0 r_2}{r_1 - r_2} = \frac{C_0 r_1 - C_1}{r_1 - r_2}$$

$$r_1 \neq r_2$$



Example 3

- Solve the recurrence $a_n = a_{n-1} + 2a_{n-2}$ given the initial conditions $a_0 = 2, a_1 = 7$.
- Solution: Use theorem 1
 - $c_1 = 1, c_2 = 2$
 - Characteristic equation:
$$r^2 - r - 2 = 0$$
 - Solutions: $r = [-(-1) \pm ((-1)^2 - 4 \cdot 1 \cdot (-2))^{1/2}] / 2 \cdot 1$
 $= (1 \pm 9^{1/2}) / 2 = (1 \pm 3) / 2,$
 - $r_1 = 2$ or $r_2 = -1$.
 - So $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$.

Example 3 Continued...

- To find α_1 and α_2 , solve the equations for the initial conditions a_0 and a_1 :

$$a_0 = 2 = \alpha_1 2^0 + \alpha_2 (-1)^0$$

$$a_1 = 7 = \alpha_1 2^1 + \alpha_2 (-1)^1$$

Simplifying, we have the pair of equations:

$$2 = \alpha_1 + \alpha_2$$

$$7 = 2\alpha_1 - \alpha_2$$

which we can solve easily by substitution:

$$\alpha_2 = 2 - \alpha_1; \quad 7 = 2\alpha_1 - (2 - \alpha_1) = 3\alpha_1 - 2;$$

$$9 = 3\alpha_1;$$

$$\alpha_1 = 3; \quad \alpha_2 = -1.$$

- Final answer: $a_n = 3 \cdot 2^n - (-1)^n$

Check: $\{a_{n \geq 0}\} = 2, 7, 11, 25, 47, 97 \dots$

Example 4

- Fibonacci sequence
- $f_n = f_{n-1} + f_{n-2}$ and $f_1 = f_2 = 1$

- $r^2 - r - 1 = 0$. $r_1 = \frac{1+\sqrt{5}}{2}$ $r_2 = \frac{1-\sqrt{5}}{2}$

- $f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$, 帶入 $f_1 = f_2 = 1$

- $\alpha_1 = \frac{1}{\sqrt{5}}$, $\alpha_2 = \frac{-1}{\sqrt{5}}$

- $f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n$

The Case of Degenerate Roots 有重根的情况

- Now, what if the C.E. $r^2 - c_1 r - c_2 = 0$ has only 1 root r_0 ?
- **Theorem 2:** Then,
$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n, \text{ for all } n \geq 0,$$

for some constants α_1, α_2 .

Example 5

- Find solution of recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} \text{ with initial condition } a_0=1, a_1=6$$

$$r^2 - 6r + 9 = 0, \quad \Rightarrow r = 3.$$

$$\text{Hence } a_n = \alpha_1 3^n + \alpha_2 n 3^n$$

From initial conditions,

$$a_0=1=\alpha_1,$$

$$a_1=6=\alpha_1 * 3 + \alpha_2 * 3$$

$$\Rightarrow \alpha_1 = 1 \text{ and } \alpha_2 = 1$$

$$\text{Therefore, } a_n = 3^n + n 3^n$$

k -LiHoReCoCos -无重根

- Consider a k -LiHoReCoCo:

- It's C.E. is:

$$r^k - \sum_{i=1}^k c_i r^{k-i} = 0$$

$$a_n = \sum_{i=1}^k c_i a_{n-i}$$

- **Thm.3:** If this has k distinct roots r_i , then the solutions to the recurrence are of the form:

$$a_n = \sum_{i=1}^k \alpha_i r_i^n$$

for all $n \geq 0$, where the α_i are constants.

Example 6

- Find the solution to the recurrence relation
- $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial condition $a_0 = 2, a_1 = 5$, and $a_2 = 15$
- $r^3 - 6r^2 + 11r - 6 = 0 \Rightarrow r_1 = 1, r_2 = 2$, and $r_3 = 3$
- $a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n$.
- $a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3$, $a_1 = 5 = \alpha_1 \cdot 1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3$,
 $a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9$.
- $\alpha_1 = 1, \alpha_2 = -1$, and $\alpha_3 = 2$.
- $\Rightarrow a_n = 1 - 2^n + 2 \cdot 3^n$.

Degenerate k -LiHoReCoCos

有重根情况

- Suppose there are t roots $r_1, \dots, r_t$ with multiplicities $m_1, \dots, m_t$. Then:

$$a_n = \sum_{i=1}^t \left(\sum_{j=0}^{m_i-1} \alpha_{i,j} n^j \right) r_i^n$$

for all $n \geq 0$, where all the α are constants.

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

Example 8

- Find the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with the initial condition $a_0 = 1, a_1 = -2$, and $a_2 = -1$.

- Solution: $r^3 + 3r^2 + 3r + 1 = 0. \Rightarrow r = -1$

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2) (-1)^n$$
$$= \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n.$$

$$a_0 = 1 = \alpha_{1,0}$$

$$a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2},$$

$$a_2 = -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}.$$

$$\alpha_{1,0} = 1, \alpha_{1,1} = 3, \text{ and } \alpha_{1,2} = -2.$$

$$\Rightarrow a_n = (1 + 3n - 2n^2)(-1)^n.$$

LiNoReCoCos

线性非齐次递推关系求解

- Linear nonhomogeneous RRs with constant coefficients may (unlike LiHoReCoCos) contain some terms $F(n)$ that depend *only* on n (and *not* on any a_i 's). General form:

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$

The *associated homogeneous recurrence relation* (associated LiHoReCoCo).

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients (*cont.*)

The following are linear nonhomogeneous recurrence relations with constant coefficients: $a_n = a_{n-1} + 2^n$,

$$a_n = a_{n-1} + a_{n-2} + n^2 + n + 1,$$

$$a_n = 3a_{n-1} + n3^n,$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$$

where the following are the associated linear homogeneous recurrence relations, respectively:

$$a_n = a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2},$$

$$a_n = 3a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

Solutions of LiNoReCoCos

常系数线性非齐次递推关系

- A useful theorem about LiNoReCoCos:
 - If $a_n = p(n)$ is any **particular solution** (特解) to the LiNoReCoCo

$$a_n = \left(\sum_{i=1}^k c_i a_{n-i} \right) + F(n)$$

- Then *all* its solutions are of the form:

$$a_n = p(n) + h(n),$$

where $a_n = h(n)$ is any solution to the associated homogeneous RR

$$a_n = \left(\sum_{i=1}^k c_i a_{n-i} \right)$$

Proof: Because $\{a_n^{(p)}\}$ is a particular solution of the nonhomogeneous recurrence relation, we know that

$$a_n^{(p)} = c_1 a_{n-1}^{(p)} + c_2 a_{n-2}^{(p)} + \cdots + c_k a_{n-k}^{(p)} + F(n).$$

Now suppose that $\{b_n\}$ is a second solution of the nonhomogeneous recurrence relation, so that

$$b_n = c_1 b_{n-1} + c_2 b_{n-2} + \cdots + c_k b_{n-k} + F(n).$$

Subtracting the first of these two equations from the second shows that

$$b_n - a_n^{(p)} = c_1 (b_{n-1} - a_{n-1}^{(p)}) + c_2 (b_{n-2} - a_{n-2}^{(p)}) + \cdots + c_k (b_{n-k} - a_{n-k}^{(p)}).$$

It follows that $\{b_n - a_n^{(p)}\}$ is a solution of the associated homogeneous linear recurrence, say, $\{a_n^{(h)}\}$. Consequently, $b_n = a_n^{(p)} + a_n^{(h)}$ for all n . 

Example 9

- Find all solutions to $a_n = 3a_{n-1} + 2n$. Which solution has $a_1 = 3$?
 - Its associated 1-LiHoReCoCo is $a_n = 3a_{n-1}$, whose solutions are all of the form $h(n) = \alpha 3^n$.
 - Thus the solutions to the original problem are all of the form $a_n = p(n) + \alpha 3^n$. So, all we need to do is find one $p(n)$ that works.

Example 9

$$a_n = 3a_{n-1} + 2n$$

- If the extra terms $F(n)$ are a degree- t polynomial in n , you should try a degree- t polynomial as the particular solution $p(n)$.
- This case: $F(n)$ is linear so try
 - $a_n = cn + d$.
$$cn + d = 3(c(n-1) + d) + 2n \quad (\text{for all } n)$$
$$(-2c - 2)n + (3c - 2d) = 0 \quad (\text{collect terms})$$
$$\text{So } c = -1 \text{ and } d = -3/2.$$
$$\text{So } a_n = -n - 3/2 \text{ is a solution.}$$
- Check: $a_{n \geq 1} = \{-5/2, -7/2, -9/2, \dots\}$

Example 9

Finding a Desired Solution

- From the previous, we know that all general solutions to our example are of the form:

$$a_n = -n - 3/2 + \alpha 3^n.$$

Solve this for α for the given case,

$$a_1 = 3:$$

$$3 = -1 - 3/2 + \alpha 3^1$$

$$\alpha = 11/6$$

- The answer is $a_n = -n - 3/2 + (11/6)3^n$

Example- LiNoReCoCos

例： Hanoi塔

- $T(n) = 2T(n-1) + 1$
- 特解设为P, 代入上式, $P = 2P + 1$,
- 所以 $P = -1$
- 齐次方程解 $h(n)$: $r = 2$, 所以 $T(n) = c \cdot 2^n - 1$, 代入初值: $T(1) = 1$, 得 $c = 1$
- 所以 $T(n) = 2^n - 1$

换元法求解

思路：通过换元转化成常系数线性递推方程

例
$$\begin{cases} a_n^2 = 2a_{n-1}^2 + 1 \\ a_0 = 2 \end{cases} \quad a_n > 0$$

令 $b_n = a_n^2$, 代入得

$$b_n = 2b_{n-1} + 1, \quad b_0 = 4$$

解得：特解 $P = -1$, 通解 $h(n) = 5 \cdot 2^n$

$$b_n = 5 \cdot 2^n - 1,$$

$$a_n = \sqrt{5 \cdot 2^n - 1}.$$

Homework

- §8.2
 - 10, 42, 46

§8.3: Divide & Conquer R.R.s

分治算法和递推关系

Main points so far:

- Many types of problems are solvable by reducing a problem of size n into some number a of **independent** subproblems, each of size $\leq \lceil n/b \rceil$, where $a \geq 1$ and $b > 1$.
- The time complexity to solve such problems is given by a recurrence relation:

$$T(n) = a T(\lceil n/b \rceil) + g(n)$$

Time for each subproblem

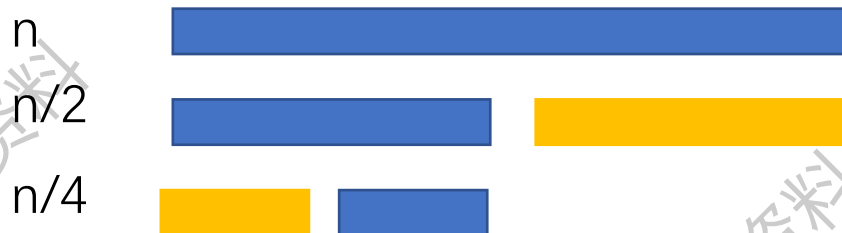
Time to break problem up into subproblems

Divide+Conquer Examples

- **example1.**

- **Binary search (二分搜索)** : Break list into 1 sub-problem (smaller list) (so $a=1$) of size $\leq \lceil n/2 \rceil$ (so $b=2$).

- So $T(n) = T(\lceil n/2 \rceil) + c$ ($g(n)=c$ constant)



Divide+Conquer Examples

• Example 3.

• Merge sort (归并排序) :

Break list of length n into 2 sublists ($a=2$), each of size $\leq \lceil n/2 \rceil$ (so $b=2$), then merge them, in

• $g(n) = \Theta(n)$ time.

- So $T(n) = 2T(\lceil n/2 \rceil) + cn$ (roughly, for some c)

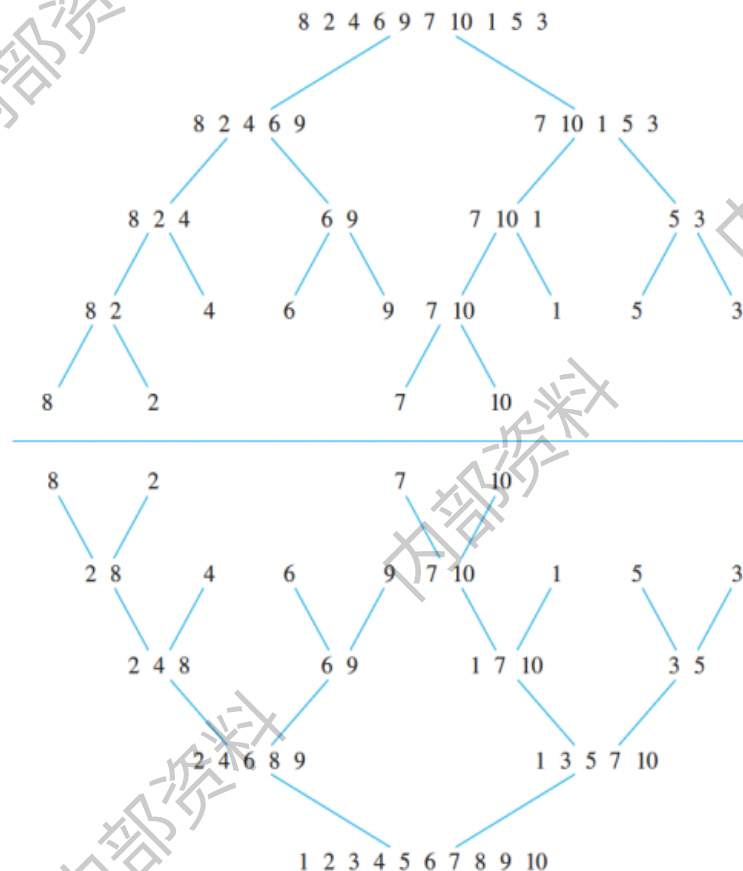


FIGURE 2 The merge sort of 8, 2, 4, 6, 9, 7, 10, 1, 5, 3.

Fast Multiplication Example

- The ordinary grade-school algorithm takes $\Theta(n^2)$ steps to multiply two n -digit numbers.
 - This seems like too much work!
- So, let's find an asymptotically *faster* multiplication algorithm!
- To find the product cd of two $2n$ -digit base- b numbers,
- $c = (c_{2n-1}c_{2n-2}\dots c_0)_b$ and $d = (d_{2n-1}d_{2n-2}\dots d_0)_b$,
 - break c and d in half:
 $c = b^n C_1 + C_0$, $d = b^n D_1 + D_0$,
 $C_1 = (c_{2n-1}\dots c_{n+1}c_n)_2$, $C_0 = (c_{n-1}\dots c_1c_0)_2$,
 $D_1 = (d_{2n-1}\dots d_{n+1}d_n)_2$, $D_0 = (d_{n-1}\dots d_1d_0)_2$.
 - and then... (see next slide)

$n=2$ 时, a, b 都是2位数: 比如 $a=38=3 \times 10+8$; $b=56=5 \times 10+6$

$$a=p \times 10+q, b=r \times 10+s$$

$$a \times b = (p \times 10+q) \times (r \times 10+s)$$

$$= (\text{purple } p \times r) \times 100 + (\text{red } p \times s + \text{red } q \times r) \times 10 + \text{green } q \times s$$

必须计算的

必须计算的

$$O(n^{\log 4}) = O(n^2)$$

(4次乘法)

$$(u+w-v) \times 10$$

$$v = (q-p) \times (s-r)$$

$$= \text{purple } p \times r + \text{green } q \times s -$$

$$(\text{red } p \times s + \text{red } q \times r)$$

$$O(n^{\log 3})$$

(3次乘法)

$$= (\text{purple } p \times r) \times 100 + (\text{purple } p \times r + \text{green } q \times s - (\text{red } q-p) \times (\text{red } s-r)) \times 10 +$$

$$\text{green } q \times s$$

$$u = \text{purple } p \times r, w = \text{green } q \times s,$$

$$(\text{red } p \times s + \text{red } q \times r) = u + w - v$$

$n=100$ 时

$$O(n^2) = 10000$$

$$O(n^{\log 3}) = 1480$$

Derivation of Fast Multiplication

$$\begin{aligned}
 cd &= (b^n C_1 + C_0)(b^n D_1 + D_0) \\
 &= b^{2n} C_1 D_1 + b^n (C_1 D_0 + C_0 D_1) + C_0 D_0 \quad \text{(Multiply out polynomials)} \\
 &= b^{2n} C_1 D_1 + C_0 D_0 + \\
 &\quad b^n (C_1 D_0 + C_0 D_1 + \underbrace{(C_1 D_1 - C_1 D_1)}_{\text{Zero}} + \underbrace{(C_0 D_0 - C_0 D_0)}_{\text{Zero}}) \\
 &= (b^{2n} + b^n) \underbrace{C_1 D_1}_{\text{Zero}} + (b^n + 1) \underbrace{C_0 D_0}_{\text{Zero}} + \\
 &\quad b^n (C_1 D_0 - C_1 D_1 - C_0 D_0 + C_0 D_1) \\
 &= (b^{2n} + b^n) \underbrace{C_1 D_1}_{\text{Zero}} + (b^n + 1) \underbrace{C_0 D_0}_{\text{Zero}} + \\
 &\quad b^n (C_1 - C_0)(D_0 - D_1) \quad \text{(Factor last polynomial)}
 \end{aligned}$$

Three multiplications, each with n -digit numbers

Recurrence Rel. for Fast Mult.

Notice that the time complexity $T(n)$ of the fast multiplication algorithm obeys the recurrence:

- $T(2n) = 3T(n) + \Theta(n)$

i.e.,

- $T(n) = 3T(n/2) + \Theta(n)$

So $a=3$, $b=2$.

Time to do the needed adds & subtracts of n -digit and $2n$ -digit numbers

Example 5

- Fast Matrix Multiplication
- Seven multiplication of two $(n/2) \times (n/2)$ matrices and 15 additions of $(n/2) \times (n/2)$ matrices.
- $f(n) = 7f(n/2) + 15n^2/4$

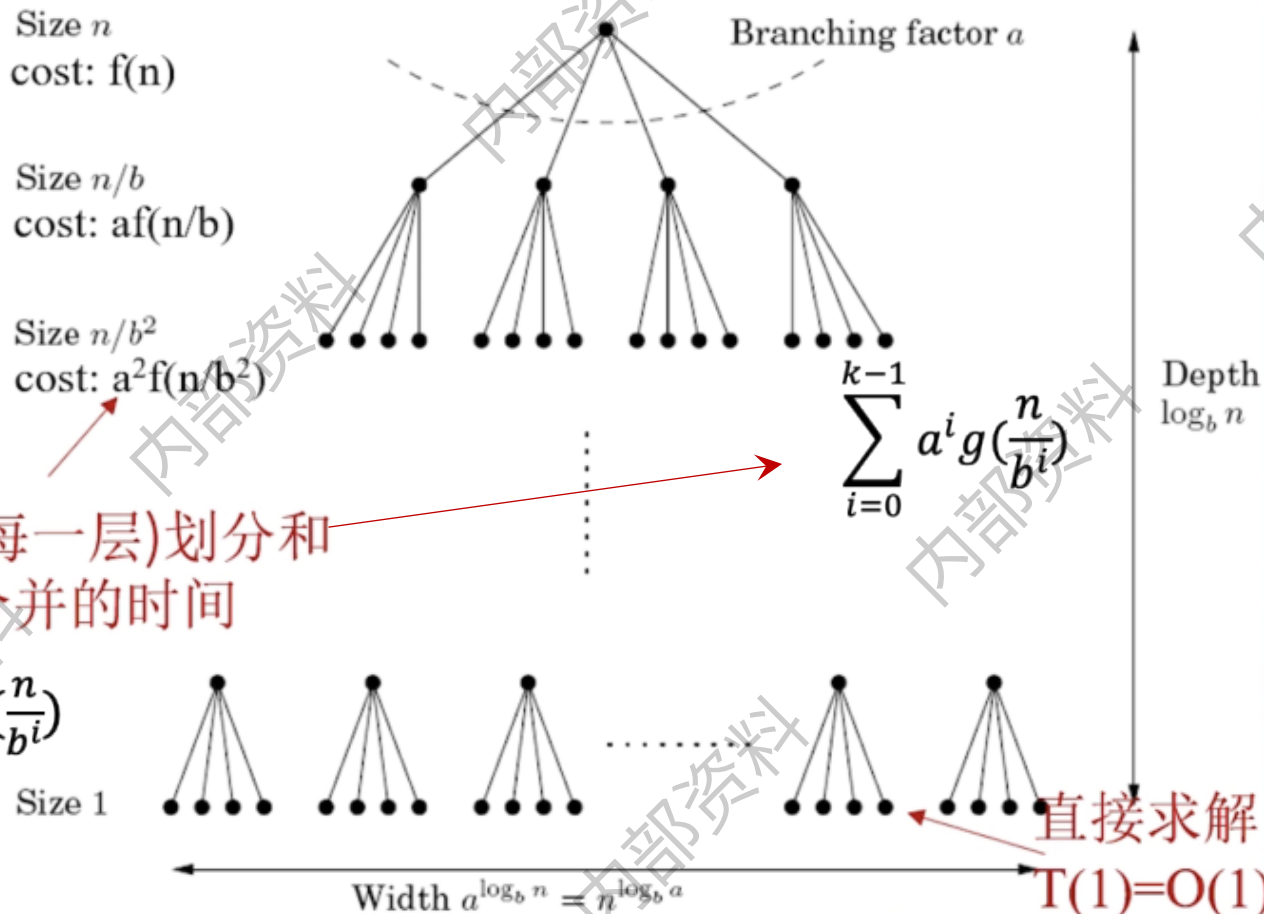


- Let $n=b^k$, where k is a positive integer. then
- $f(n)=af(n/b)+g(n)$
 - $=a^2f(n/b^2)+ag(n/b)+g(n)$
 - $=a^3f(n/b^3)+a^2f(n/b^2)+ag(n/b)+g(n)$
 -
 - $=a^kf(\frac{n}{b^k})+\sum_{i=0}^{k-1}a^ig(\frac{n}{b^i})$
 - $=a^kf(1)+\sum_{i=0}^{k-1}a^ig(\frac{n}{b^i})$



递归树

Let $n = b^k$



$$f(n) = a^k f(1) + \sum_{i=0}^{k-1} a^i g\left(\frac{n}{b^i}\right)$$



$$f(n) = a^k f(1) + \sum_{i=0}^{k-1} a^i g\left(\frac{n}{b^i}\right)$$

当 $g\left(\frac{n}{b^i}\right) = c$ (常数)

$$f(n) = \begin{cases} f(1) + ck = O(kc) = O(\log n), & a = 1 \\ a^k f(1) + c \frac{a^k - 1}{a - 1} = O(a^k) = O(n^{\log_b a}), & a \neq 1 \end{cases}$$

$$n = b^k, \quad k = \log_b n, \quad a^k = a^{\log_b n}$$



Theorem 1

Let f be an increasing function that satisfies the recurrence relation:

$$f(n) = af(n/b) + c$$

with $a \geq 1$, integer $b > 1$, real $c > 0$, Then:

$$f(n) \text{ is } \begin{cases} O(n^{\log_b a}) & \text{if } a > 1 \\ O(\log n) & \text{if } a = 1 \end{cases}$$

when $n=b^k$, $a \neq 1$, for all $k \in \mathbb{Z}^+$,

$$f(n) = C_1 n^{\log_b a} + C_2$$

Where $C_1 = f(1) + c/(a-1)$ and $C_2 = -c(a-1)$

proof (1)

• let $n=b^k$,
$$f(n) = a^k f(1) + \sum_{j=0}^{k-1} a^j c = a^k f(1) + c \sum_{j=0}^{k-1} a^j.$$

when $a=1$,

and $k=\log_b n$. $f(n)=f(1)+c \log_b n.$

$$\begin{aligned} f(n) &\leq f(b^{k+1}) = f(1) + c(k+1) = (f(1) + c) + ck \\ &\leq (f(1) + c) + c \log_b n. \end{aligned}$$

($n=b^k$)
($b^k < n < b^{k+1}$)

Therefore, $f(n)$ is $O(\log n)$ when $a=1$.

proof (2)

$$f(n) = a^k f(1) + \sum_{j=0}^{k-1} a^j c = a^k f(1) + c \sum_{j=0}^{k-1} a^j.$$

when $a > 1$, and $k = \log_b n$.

$$f(n) = a^k f(1) + c(a^k - 1)/(a - 1)$$

$$= a^k [f(1) + c/(a - 1)] - c/(a - 1)$$

$$= C_1 n^{\log_b a} + C_2 \quad \text{because } a^k = a^{\log_b n} = n^{\log_b a}$$

$$f(n) \leq f(b^{k+1}) = C_1 a^{k+1} + C_2$$

$$\leq (C_1 a) a^k + C_2 \quad (k \leq \log_b n \leq k+1)$$

$$= (C_1 a) n^{\log_b a} + C_2$$

Therefore, $f(n)$ is $O(n^{\log_b a})$ when $a > 1$.

Example

➤ Example 6:

- Let $f(n)=5f(n/2)+3$ and $f(1)=7$. Find $f(2^k)$
- where k is a positive integer. Also, estimate $f(n)$ if f is an increasing function.
- $a=5, b=2, c=3, O(n^{\log a}) = O(n^{\log 5})$

➤ Example 7:

- Binary search: $f(n)=f(n/2)+2$
- $a=1, b=2, c=2, O(\log(n))$

The Master Theorem 主定理

Consider a function $f(n)$ that, for all $n=b^k$ for all $k \in \mathbf{Z}^+$, satisfies the recurrence relation:

$$f(n) = af(n/b) + cn^d$$

with $a \geq 1$, integer $b > 1$, real $c > 0$, $d \geq 0$. Then:

$$f(n) \in \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d \log n) & \text{if } a = b^d \\ O(n^{\log_b a}) & \text{if } a > b^d \end{cases}$$

Example 9

- Complexity of Merge sort (归并排序)
- $M(n) = 2M(n/2) + n$
- Thus, $a=2$, $b=2$, $d=1$. So $a = b^d$, so case 2 of the master theorem applies, so:
- $M(n)$ is $O(n \log n)$.

Recurrences

- $T(1) = 0$
- $T(n) = 2 * T(n/2) + n$
 - $a=2, b=2, d=1; a=b^d, T(n)=O(n \log_2 n)$
 - Check :
 - $$\begin{aligned} T(n) &= 2 T(n/2) + n \\ &= 2 (n/2 \log_2(n/2)) + n \\ &= n \log_2(n/2) + n \\ &= n \log_2(n) - n \log_2 2 + n \\ &= n \log_2(n) - n + n \\ &= n \log_2(n) \end{aligned}$$

Recurrence relations

方法二

$$T(1) = 0$$

$$T(n) = 2 * T(n/2) + n$$

$$n = 2^k \quad k = 0, 1, 2, \dots, \quad (k = \log_2 n)$$

$$S(k) = 2 * S(k-1) + 2^k \quad S(0) = 0, S(1) = 2$$

$$S(k-1) = 2 * S(k-2) + 2^{k-1}$$

$$S(k) = 4 * S(k-1) - 4 * S(k-2)$$

$$S(k) = u2^k + vk2^k$$

$$u = 0, v = 1 \quad S(k) = k2^k,$$

$$\Rightarrow T(n) = n \log_2(n)$$

Master Theorem Example

- Recall that complexity of fast multiply was:

$$T(n) = 3T(n/2) + \Theta(n)$$

- Thus, $a=3$, $b=2$, $d=1$. So $a > b^d$, so case 3 of the master theorem applies, so:

$$T(n) = O(n^{\log_b a}) = O(n^{\log_2 3})$$

which is $O(n^{1.58\dots})$, so the new algorithm is strictly faster than ordinary $\Theta(n^2)$ multiply!

Example 11

- Fast Matrix Multiplication
- Seven multiplication of two $(n/2) \times (n/2)$ matrices and 15 additions of $(n/2) \times (n/2)$ matrices.
- $f(n) = 7f(n/2) + 15n^2/4$

$a=7, b=2, d=2, \text{ So } a > b^d,$

so: $f(n) = O(n^{\log_2 7})$

8.4 Generating Functions.

Def 1. The **generating function** for the sequence $\{a_n\}$ is the **infinite** series.

$$G(x) = a_0 + a_1x + \dots + a_nx^n + \dots$$

$$= \sum_{k=0}^{\infty} a_k x^k$$

finite sequences of real numbers, $a_{n+1} = 0, a_{n+2} = 0,$

and so on

$$G(x) = a_0 + a_1x + \dots + a_nx^n$$

$$A(x) = \sum_{n=0}^{\infty} a(n) \cdot \frac{x^n}{n!}$$

$$a_n = 1, \quad A(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

Example 2. What is the generating function for the sequence 1,1,1,1,1,1 ?

Sol :

$$G(x) = 1 + x + x^2 + \dots + x^5 \quad (\text{expansion})$$

$$= \frac{x^6 - 1}{x - 1} \quad (\text{closed form})$$

Example 3.

Let $m \in \mathbf{Z}^+$ and $a_k = \binom{m}{k}$, for $k = 0, 1, \dots, m$.

What is the generating function for the sequence $a_0, a_1, \dots, a_m$?

Sol :

$$\begin{aligned} G(x) &= a_0 + a_1x + a_2x^2 + \dots + a_mx^m \\ &= \binom{m}{0} + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m \end{aligned}$$

$$= (1+x)^m \quad (\text{by 二项式定理})$$

Example 4.

The function $f(x) = \frac{1}{1-x}$ is the generating

function of the sequence 1, 1, 1, 1, ...,

$$\text{since } \frac{1}{1-x} = 3(1 + z/3 + (z/3)^2 + \dots) = 3 \sum_{n=0}^{\infty} \left(\frac{z}{3}\right)^n$$

when $|z/3| < 1$.

Example 5.

The function $f(x) = \frac{1}{1-ax}$ is the generating

function of the sequence $1, a, a^2, \dots$,

$$\text{since } \frac{1}{1-ax} = 1 + ax + a^2x^2 + \dots = \sum_{k=0}^{\infty} (ax)^k$$

when $|ax| < 1$ for $a \neq 0$

Theroem1

• Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$. Then

$$f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k$$

• and

$$f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

$$\begin{aligned} & (a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + \cdots) (b_0 + b_1 x + b_2 x^2 + \cdots + b_k x^k + \cdots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0) x + (a_1 b_1 + a_2 b_0 + a_0 b_2) x^2 + (a_2 b_1 + a_1 b_2 + a_3 b_0 + a_0 b_3) x^3 + \cdots \end{aligned}$$

Example 6

Let $f(x)=1/(1-x)^2$. Use Example 4

$$\frac{1}{(1-x)} = 1+x+x^2+x^3+....$$

to find the coefficients $a_0, a_1, a_2, \dots$ in the expansion

$$\begin{aligned}\frac{1}{(1-x)^2} &= \sum_{k=0}^{\infty} \left(\sum_{j=0}^k 1 \right) x^k = \sum_{k=0}^{\infty} (k+1) x^k \\ &= 1+2x+3x^2+....\end{aligned}$$

Definition 2. 广义二项式系数

Let $u \in \mathbf{R}$ and $k \in \mathbf{Z}^+ \cup \{0\}$. Then the extended

binomial coefficient $\binom{u}{k}$ is defined by

$$\binom{u}{k} = \begin{cases} \frac{1}{k!} u(u-1)(u-2)\dots(u-k+1), & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$$

Example 7.

Find $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} \frac{1}{2} \\ 3 \end{pmatrix}$

Sol :

$$\begin{pmatrix} -2 \\ 3 \end{pmatrix} = \frac{1}{3!}(-2)(-3)(-4) = -4$$

$$\begin{pmatrix} \frac{1}{2} \\ 3 \end{pmatrix} = \frac{1}{3!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right) = \frac{1}{16}$$

The extended binomial theorem

广义二项式定理

- Theorem 2

- Let $x \in \mathbf{R}$ with $|x| < 1$ and let $u \in \mathbf{R}$. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

- Table 1 Useful Generating Functions

Example 9.

Find the generating functions for $(1+x)^{-n}$ and $(1-x)^{-n}$ where $n \in \mathbf{Z}^+$

Sol : By the Extended Binomial Theorem,

$$\begin{aligned}(1+x)^{-n} &= \sum_{k=0}^{\infty} \binom{-n}{k} x^k = \sum_{k=0}^{\infty} \frac{1}{k!} (-n)(-n-1)\dots(-n-k+1)x^k \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} (n)(n+1)\dots(n+k-1)x^k \\ &= \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k\end{aligned}$$

Note: $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$

By replacing x by $-x$ we have

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$$

有用的生成函数

The following table lists a number of useful generating functions, many of which can be obtained from known formulas for sums of series.

Useful Generating Functions	
$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$	$C(n, k/r)$ if $r k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk}$	1 if $r k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$	$C(n+k-1, k)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$	$(-1)^k C(n+k-1, k)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$	$C(n+k-1, k)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$	$1/k!$
$\ln(1+x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{k} x^k$	$(-1)^{k+1}/k$

※ Counting Problems and Generating Functions

Example 10.

Find the number of solutions of

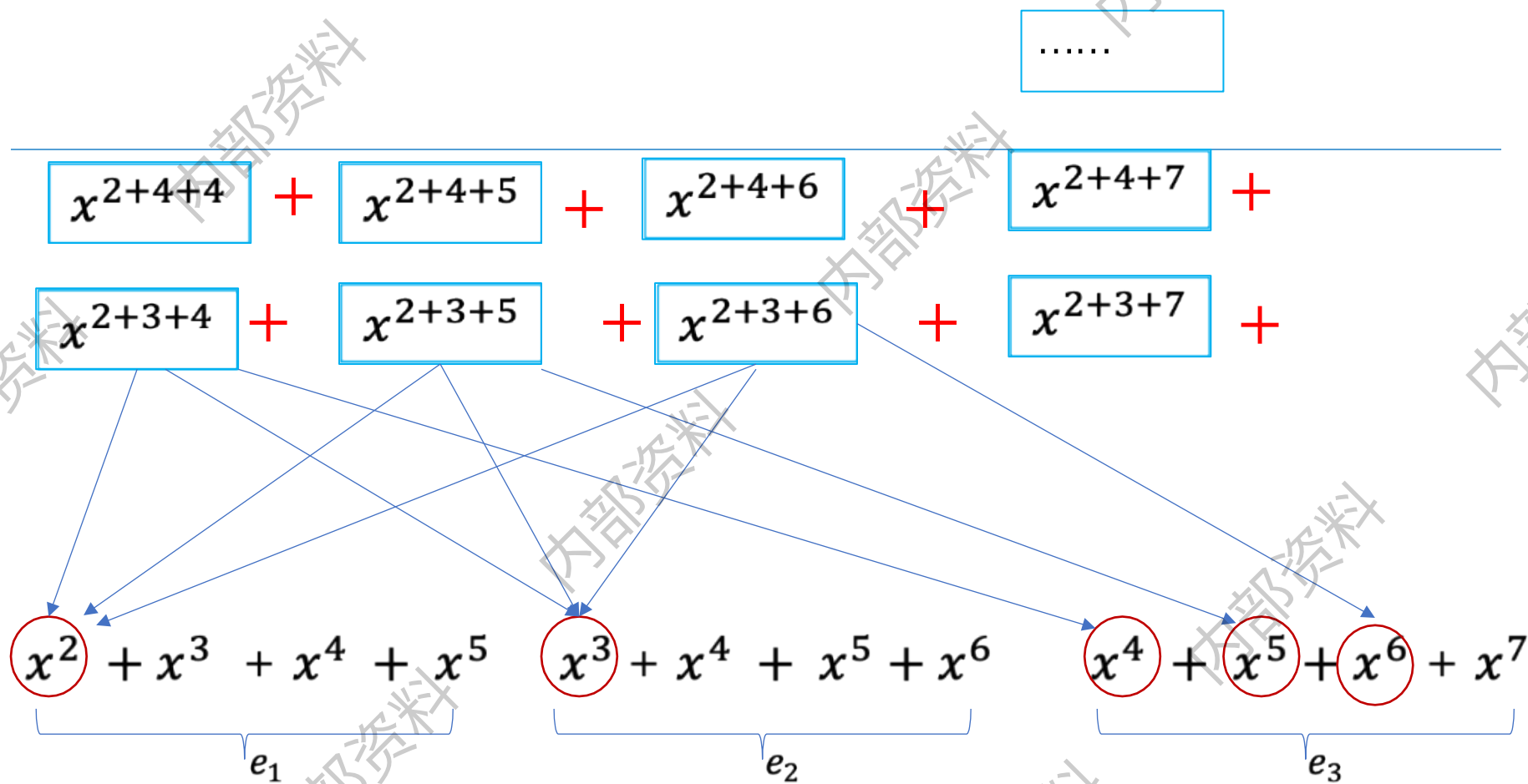
$$e_1 + e_2 + e_3 = 17$$

Where e_1, e_2 and e_3 are nonnegative integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$ and $4 \leq e_3 \leq 7$.

Solution:

The number of solutions is the coefficient of x^{17} in the expansion of

$$x^{e_1} x^{e_2} x^{e_3} = x^{e_1 + e_2 + e_3} = x^{17}$$



※ Counting Problems and Generating Functions

Example 10.

$$x^{e1}x^{e2}x^{e3}=x^{e1+e2+e3}=x^{17}$$

$$\begin{aligned} & (x^2 + x^3 + x^4 + x^5) (x^3 + x^4 + x^5 + x^6) (x^4 + x^5 + x^6 + x^7) \\ &= x^2x^3x^4 + x^2x^3x^5 + x^2x^3x^6 + x^2x^3x^7 + \\ & \quad \dots \dots + x^4x^6x^7 + x^5x^5x^7 + x^5x^6x^6 \\ & \quad \quad \quad + \dots \dots \end{aligned}$$

$$= x^9 + \dots \dots + 3x^{17} + \dots \dots + x^{18}$$

There are three solutions since the coefficient of x^{17} in the product is 3.

※ Counting Problems and Generating Functions

Example 10-1.

Find the number of solutions of

$$e_1 + e_2 + e_3 = 17$$

Where e_1, e_2 and e_3 are nonnegative integers

Solution:

$$x^{e_1} x^{e_2} x^{e_3} = x^{e_1 + e_2 + e_3} = x^{17}$$

$$(1 + x^1 + x^2 + \cdots) (1 + x^1 + x^2 + \cdots) (1 + x^1 + x^2 + \cdots) \\ = 1 + 3x + 6x^2 + \cdots \cdots + 171x^{17} + \cdots \cdots$$

$$G(x) = \frac{1}{(1-x)} = 1+x+x^2+x^3+\dots$$

$$G(x) \cdot G(x) = \frac{1}{(1-x)^2} = G(x)' = 1+2x+3x^2+4x^3+\dots$$

$$\begin{aligned} G(x) \cdot G(x) \cdot G(x) &= \frac{1}{(1-x)^3} = \frac{1}{2} G(x)'' \\ &= \frac{1}{2}(2+3 \cdot 2x+4 \cdot 3x^2+5 \cdot 4x^3+\dots+19 \cdot 18x^{17}+\dots+n \cdot (n-1)x^{n-2}) \end{aligned}$$

利用3个生成函数的乘积得到新的生成函数

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$$

Note: $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$

Example 12

- Use generating functions to determine the number of ways to insert **tokens worth \$1, \$2, and \$5** into a vending machine to pay for an item that costs r dollars in both the cases when the order in which the tokens are inserted **does not matter**. (不考虑插入顺序) and when **the order does matter**.

• *Solution 1:*

➤ $(1+x+x^2+x^3+\dots)(1+x^2+x^4+x^6+\dots)(1+x^5+x^{10}+x^{15}+\dots).$

- $r=7$, $result=6$

Solution 2: $(x+x^2+x^5)^n$

$$1 + (x+x^2+x^5) + (x+x^2+x^5)^2 + \dots = \frac{1}{1 - (x+x^2+x^5)}$$

Example 14

- Use generating functions to find the number of
- **r-combinations** (r-组合数) from a set with n elements when **repetition of elements** is allowed.
- solution: $G(x) = (1+x+x^2+\cdots)^n$.
- As long as $|x| < 1$, we have $1+x+x^2+\cdots=1/(1-x)$, so $G(x) = 1/(1-x)^n = (1-x)^{-n}$.
- $(1-x)^{-n} = (1+(-x))^{-n} = \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r$
- $a_r = \binom{-n}{r} (-1)^r = (-1)^r C(n+r-1, r) \cdot (-1)^r = C(n+r-1, r)$.

Example 15

- Use generating functions to find the number of ways to select r objects of n different kinds if we must select **at least one object** of each kind.

$$G(x) = (x + x^2 + x^3 + \cdots)^n = x^n(1 + x + x^2 + \cdots)^n = x^n/(1 - x)^n.$$

$$\begin{aligned} G(x) &= x^n/(1 - x)^n \\ &= x^n \cdot (1 - x)^{-n} \\ &= x^n \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r \\ &= x^n \sum_{r=0}^{\infty} (-1)^r C(n + r - 1, r) (-1)^r x^r \\ &= \sum_{r=0}^{\infty} C(n + r - 1, r) x^{n+r} \\ &= \sum_{t=n}^{\infty} C(t - 1, t - n) x^t \\ &= \sum_{r=n}^{\infty} C(r - 1, r - n) x^r. \end{aligned}$$

※ Using Generating Functions to solve Recurrence Relations. 生成函数求解递推方程

Example 16.

Solving the recurrence relation $a_k = 3a_{k-1}$ for $k=1,2,3,\dots$ and initial condition $a_0 = 2$.

Sol :

另法 :

$$r - 3 = 0 \Rightarrow r = 3 \Rightarrow a_n = \alpha \cdot 3^n$$

$$\therefore a_0 = 2 = \alpha$$

$$\therefore a_n = 2 \cdot 3^n$$

Example 6 生成函数求解

Let $G(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{k=0}^{\infty} a_k x^k$ be the generating function for $\{a_k\}$.

First note that $a_k x^k = 3a_{k-1} x^k$

$$\Rightarrow \sum_{k=1}^{\infty} a_k x^k = 3 \sum_{k=1}^{\infty} a_{k-1} x^k = 3x \sum_{k=1}^{\infty} a_{k-1} x^{k-1} = 3x \sum_{k=0}^{\infty} a_k x^k$$

$$\Rightarrow G(x) - a_0 = 3x \cdot G(x)$$

$$\because a_0 = 2 \Rightarrow G(x) - 3x \cdot G(x) = G(x)(1-3x) = 2$$

$$\therefore G(x) = \frac{2}{1-3x} = 2 \cdot \sum_{k=0}^{\infty} (3x)^k = \sum_{k=0}^{\infty} 2 \cdot 3^k \cdot x^k$$

$$\therefore a_k = 2 \cdot 3^k$$

※ Using Generating Functions to solve Recurrence Relations.

Example 17.

Suppose that a valid codeword is an n -digit number in decimal notation containing an even number of 0s. Let a_n denote the number of valid codewords of length n . Use generating functions to find an explicit formula for a_n .

Solving the recurrence relation

$$a_n = 8a_{n-1} + 10^{n-1}$$

for $n=1,2,3,\dots$ and initial condition $a_1 = 9$.



$$a_n x^n = 8a_{n-1} x^n + 10^{n-1} \cdot x^n$$

$$G(x) - 1 = \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} x^n + 10^{n-1} x^n)$$

$$= 8 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n$$

$$= 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1}$$

$$= 8x \sum_{n=1}^{\infty} a_n x^n + x \sum_{n=1}^{\infty} 10^n x^n$$





$$G(x)-1=8xG(x)+x/(1-10x).$$

$$G(x)=\frac{1-9x}{(1-8x)(1-10x)}$$

$$G(x)=\frac{1}{2}\left(\frac{1}{1-8x}+\frac{1}{1-10x}\right)$$

$$G(x)=\frac{1}{2}\left(\sum_{n=0}^{\infty}8^n x^n+\sum_{n=0}^{\infty}10^n x^n\right)$$

$$=\sum_{n=0}^{\infty}\frac{1}{2}(8^n+10^n)x^n$$

$$a_n=\frac{1}{2}(8^n+10^n)$$



Proving Identities via Generating Functions

- Example 18
 - Use generating functions to show that

$$\sum_{k=0}^n C(n, k)^2 = C(2n, n)$$

When n is a positive integer.

$C(2n, n)$ is the coefficient of x^n in $(1 + x)^{2n}$

$$\begin{aligned}(1+x)^{2n} &= [(1+x)^n]^2 \\ &= [C(n, 0) + C(n, 1)x + C(n, 2)x^2 + \cdots + C(n, n)x^n]^2\end{aligned}$$

The coefficient of x^n in this expression is

$$\begin{aligned}&C(n, 0)C(n, n) + C(n, 1)C(n, n-1) + C(n, 2)C(n, n-2) + \cdots \\ &+ C(n, n)C(n, 0)\end{aligned}$$

This equals $\sum_{k=0}^n C(n, k)^2$, because $C(n, n-k) = C(n, k)$

$$\text{so, } \sum_{k=0}^n C(n, k)^2 = C(2n, n)$$

由生成函数求解

给定序列 $\{a_n\}$ 的生成函数 $G(x)$, 求 a_n

待定系数法

例

$$G(x) = \frac{2}{1-3x+2x^2} = \frac{A}{1-x} + \frac{B}{1-2x}$$

$$\begin{cases} A + B = 2 \\ -2A - B = 0 \end{cases} \Rightarrow A = -2, B = 4$$

$$G(x) = \frac{-2}{1-x} + \frac{4}{1-2x} = \sum_{n=0}^{\infty} (-2)x^n + \sum_{n=0}^{\infty} 4(2x)^n$$

$$a_n = -2 + 4 \cdot 2^n$$

生成函数求解递推方程

例 $a_n - 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 1, \quad a_1 = -2$

$$\begin{array}{rcl}
 G(x) & = & a_0 + a_1x + a_2x^2 + a_3x^3 + \dots \\
 -5xG(x) & = & -5a_0x - 5a_1x^2 - 5a_2x^3 - \dots \\
 6x^2G(x) & = & + 6a_0x^2 + 6a_1x^3 + \dots \\
 \hline
 (1 - 5x + 6x^2)G(x) & = & a_0 + (a_1 - 5a_0)x
 \end{array}$$

$$G(x) = \frac{1-7x}{1-5x+6x^2} = \frac{5}{1-2x} - \frac{4}{1-3x}$$

$$= 5 \sum_{n=0}^{\infty} 2^n x^n - 4 \sum_{n=0}^{\infty} 3^n x^n$$

$$a_n = 5 \cdot 2^n - 4 \cdot 3^n$$

生成函数求递推方程解 (续)

$$\begin{cases} h_n = \sum_{k=0}^{n-1} h_k h_{n-k} & (n \geq 2) \\ h_1 = 1 \end{cases}$$

解：设 $\{h_n\}$ 的生成函数为 $H(x) = \sum_{n=1}^{\infty} h_n x^n$

$$\begin{aligned} H^2(x) &= \sum_{k=1}^{\infty} h_k x^k \cdot \sum_{l=1}^{\infty} h_l x^l \\ &= \sum_{n=2}^{\infty} x^n \sum_{k=1}^{n-1} h_k h_{n-k} = \sum_{n=2}^{\infty} h_n x^n \end{aligned}$$

$$= H(x) - h_1 x = H(x) - x$$

$$f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

生成函数求递推方程解 (续)

$$H^2(x) - H(x) + x = 0$$

$$H(x) = \frac{1 - (1 - 4x)^{\frac{1}{2}}}{2} = \frac{1}{2} - \frac{1}{2} (1 - 4x)^{\frac{1}{2}}$$

$$= \frac{1}{2} - \frac{1}{2} \left[1 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n 2^{2n-1}} \binom{2n-2}{n-1} (-4x)^n \right]$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n 2^{2n}} \binom{2n-2}{n-1} (-1)^n 2^{2n} x^n = \sum_{n=1}^{\infty} \frac{1}{n} \binom{2n-2}{n-1} x^n$$

$$h_n = \frac{1}{n} \binom{2n-2}{n-1}$$

Catalan数

扩展应用：

矩阵乘法：设 $A_1, A_2, \dots, A_n$ 为矩阵序列， A_i 为 $P_{i-1} \times P_i$ 阶矩阵
 $i=1, 2, \dots, n$ 确定乘法顺序使得元素相乘的总次数最少

输入：向量 $P = \langle P_0, P_1, \dots, P_n \rangle$

实例： $P = \langle 10, 100, 5, 50 \rangle$

$$A_1: 10 \times 100, \quad A_2: 100 \times 5, \quad A_3: 5 \times 50$$

$$(A_1 A_2) A_3: 10 \times 100 \times 5 + 10 \times 5 \times 50 = 7500$$

$$A_1 (A_2 A_3): 10 \times 100 \times 50 + 100 \times 5 \times 50 = 75000$$

一般算法： $n-1$ 次运算加括号方法有 $\frac{1}{n} \binom{2n-2}{n-1}$ 种，

Stirling公式
$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + O\left(\frac{1}{n}\right)\right)$$

方法数
$$W(n) = \Omega\left(\frac{1}{n} \frac{(2n-2)!}{(n-1)!(n-1)!}\right) = \Omega(2^{2n}/n^{\frac{3}{2}})$$

作业

- §8.1 – 10, 14
- §8.2 – 10, 42, 46
- §8.3 – 14, 28
- §8.4 – 8, **16, 24, 36**